

Hidden States Beat Summary Statistics: Extracting Intraday Volatility Information from a Frozen Time-Series Foundation Model¹

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Abstract

Time-series foundation models (TSFMs) are commonly evaluated in finance through scalar summaries of their predictive distribution—negative log-likelihood or predictive entropy—and on that basis have been judged nearly useless for index risk management. We show that this verdict is an artifact of the *extraction interface*, not the model. Feeding the full 512-dimensional hidden state of a frozen, zero-shot K-line foundation model into a 35K-parameter gated corrector on top of a strong HAR-type baseline raises the out-of-sample gain in one-hour-ahead index volatility forecasts by an order of magnitude: on a pre-registered, one-shot holdout year, $\Delta R^2 = +0.095 \pm 0.004$ across 100 seeds (none negative), with QLIKE improvements of 17–21% (Diebold–Mariano t up to 7.6, Clark–West t up to 16.9). The hidden state subsumes a 500-constituent cross-sectional breadth signal and, on 133 holdout days of five-minute index-option order-book data, the option market’s own implied volatility; it is essentially 100% novel relative to known volatility variables, and helps most exactly when the baseline is weakest. Its incremental value traces an inverted-U in the forecast horizon that peaks at one hour, a curve that predicts both where the forecast can be monetized and where it cannot. A same-input scratch Transformer matches the frozen pipeline only with the full nine-year training sample; with half the data the frozen representation retains 77% of its gain versus 34% for the scratch model. The economic value of pretraining is thus sample efficiency in the practically relevant data regime, not asymptotic accuracy. Direction remains unpredictable throughout and the information horizon is strictly intraday. A volatility-targeted overlay on CSI 500/1000 index futures under realistic Chinese transaction costs converts the forecast upgrade into 460–970 basis points per year relative to an identical overlay driven by the baseline forecast, although strategy-level bootstrap intervals still include zero—an explicit illustration of the gap between forecast significance and portfolio significance.

JEL codes: C45, C53, C58, G17.

Keywords: time-series foundation models, transfer learning, realized volatility, volatility-managed portfolios, index futures.

¹We thank seminar participants for helpful comments. All remaining errors are our own. Code and complete experimental logs are available from the author.

1 Introduction

Pretrained foundation models for time series promise transferable representations of temporal structure (Das et al., 2024; Ansari et al., 2024; Shi et al., 2025), yet their record in quantitative finance is mixed. A common evaluation pattern treats the TSFM as a probabilistic forecaster and extracts one or two scalars—the negative log-likelihood (NLL) of realized outcomes, or the entropy of the one-step predictive distribution—as “model-based” risk factors. In our own prior work, ten successive evaluations of such scalars on Chinese index data produced economically null results and led the program to be shut down.

This paper documents a precise reversal of that verdict. Holding the model, the data, the baseline, and the evaluation protocol fixed, we change only the *extraction interface*: instead of two scalars we use the full 512-dimensional final hidden state of the frozen decoder, mapped through a small gated linear corrector in the spirit of Dewage et al. (2026). The increment to one-hour-ahead volatility forecasts rises by an order of magnitude and survives a battery of falsification tests: a pre-registered one-shot holdout year, one hundred re-trainings from independent seeds, placebo shuffling of the hidden states, nested-model Clark and West (2007) tests, and a model confidence set (Hansen et al., 2011) that eliminates every specification without hidden states.

Three findings characterize *what* the representation knows. First, second moments only: direction is unpredictable in every configuration ($R^2 \approx 0$), symmetric tail volatility inherits the full gain, while signed tail objectives (value-at-risk, crash probability) gain little. Second, the incremental value traces an inverted-U in the forecast horizon, rising from +0.013 at five minutes to +0.084 at one hour and collapsing to +0.009 for next-day realized variance: the representation encodes a regime at a characteristic timescale of about an hour. Third, the single-index hidden state *subsumes* cross-sectional breadth: conditioning on it drives the incremental value of a breadth-entropy signal built from all 500 index constituents to zero, while the converse does not hold.

Two further experiments discipline the interpretation. A scratch Transformer trained end-to-end on the identical input window matches the frozen pipeline at full sample size (only the test segment retains a gap), but falls far behind when training data is restricted to realistic spans: with 4.5 years of data the frozen representation attains 77% of its full-sample holdout gain versus 34% for the scratch model, and with half a year both fail. The premium of pretraining is therefore *sample efficiency and cross-regime stability*, not an asymptotically unreachable representation. Second, transfer tests show that the representation is universal across indices while the corrector is family-specific: a corrector trained on two broad-cap indices transfers at full strength to a third within the family, but zero-shot transfer to style indices retains only a small fraction of the gain, which local re-fitting of the 35K-parameter corrector restores.

Finally, we take the forecast to the futures market with production-grade accounting: official

dominant-contract mapping lagged one day, within-contract returns including overnight gaps, roll costs, and the asymmetric Chinese fee schedule under which intraday close-outs cost roughly seven times more than opens. A no-trade band is what keeps the overlay alive under these costs. The forecast upgrade is worth 460–970 basis points per year against an identical overlay driven by the baseline forecast (six of six index-segment cells positive; best cell $t = 2.3$), yet strategy-level bootstrap intervals still include zero: forecast significance does not automatically become portfolio significance, and the deployment we endorse is an exposure-management overlay for accounts that already hold a net long index-futures position.

Related literature. Our paper connects three strands. The first is the emerging literature on time-series foundation models: TimesFM (Das et al., 2024), Chronos (Ansari et al., 2024), and the K-line-specialized Kronos (Shi et al., 2025) pretrain autoregressive forecasters on large corpora. Dewage et al. (2026) document that frozen TSFMs are weak in high-frequency finance unless paired with lightweight neural-classical correctors; our corrector follows their gated architecture, but our ablations reverse their attribution—with a strong baseline that already absorbs seasonality and multi-scale realized volatility, tree-based residual learners add nothing after the gated corrector. We also show the phenomenon is not model-specific: a univariate Chronos encoder carries real, though three times weaker, volatility information on the same task. The second strand is volatility forecasting. Our baseline follows the HAR tradition of Corsi (2009) with multi-scale realized components and intraday seasonality dummies; HARQ-type corrections (Bollerslev et al., 2016) add nothing at the daily horizon in our data. Losses and tests follow Patton (2011), Diebold and Mariano (1995), Clark and West (2007), and Hansen et al. (2011). The third strand is volatility-managed portfolios (Moreira and Muir, 2017): our overlay is an intraday, forecast-upgraded variant, and we show that the scaling exponent ($1/\sigma$ versus $1/\sigma^2$) is a regime bet orthogonal to forecast quality.

The remainder of the paper proceeds as follows. Section 2 describes the data. Section 3 presents the frozen representation and the corrector. Section 4 contains the baseline forecasting results and their statistical validation. Section 5 characterizes the nature of the extracted information. Section 6 asks whether pretraining is necessary. Section 7 evaluates the futures overlay under realistic costs. Section 8 summarizes robustness, and Section 9 concludes.

2 Data

Indices. We use 5-minute OHLCV-plus-amount bars for the CSI 300, CSI 500, and CSI 1000 indices, aggregated from an authoritative one-minute database, covering January 2014 through August 2026 (3,068 trading days, 48 bars per day, no missing bars). Decision points are bars 0–35 of each day, so that the forecast window remains within the trading session, yielding 324,157 index-bar observations of which 315,045 survive feature filters. Transfer experiments additionally use the

SSE 50, ChiNext, and STAR 50 indices from the same source.

Futures. The economic application uses IC (CSI 500) and IM (CSI 1000) index futures. We construct dominant-contract series from the official dominant map lagged by one day (so the contract held on day t is known at the close of $t - 1$), compute within-contract 5-minute returns plus same-contract overnight returns, and account for rolls explicitly (135 rolls for IC over 2015–2026).

Sample splits. The training sample ends 2023-04-27; the validation sample (used for early stopping and *all* selection) ends 2024-06-07; the test sample (reporting only) ends 2025-07-17; and the holdout sample covers 2025-07-18 through 2026-08-14 (262 days) and is *pre-registered and opened exactly once* after all modeling decisions were fixed.

Target. The forecast objective is $y_t = \log \left(\frac{1}{H} \sum_{i=1}^H |r_{t+i}| \right)$ with $H = 12$ (one hour) or $H = 24$ (two hours), where r denotes 5-minute log returns and overnight bars are excluded. A directional control target $\sum_i r_{t+i}$ is carried through every experiment.

3 Methodology

3.1 Frozen representation

Each decision point feeds the frozen Kronos-small decoder a window of $L = 128$ bars, z-scored *within the window* and clipped at ± 5 :

$$\tilde{x}_{t-i} = \text{clip}((x_{t-i} - \mu_W)/(\sigma_W + \varepsilon), -5, 5), \quad i = 0, \dots, L - 1. \quad (1)$$

Window-local normalization forces one independent forward pass per decision point; sliding-window reuse would leak future bars into the normalizer. From the forward pass we keep the final hidden state $\mathbf{h}_t = \text{ctx}[:, L-1, :] \in \mathbb{R}^{512}$ and, as the scalar-interface control, the NLL and the one-step predictive entropy. The model is never fine-tuned.

3.2 Baseline and corrector

The baseline is a deliberately strong ridge regression,

$$\hat{y}^0 = \mathbf{x}_b^\top \hat{\beta}, \quad \hat{\beta} = (X_{\text{tr}}^\top X_{\text{tr}} + \lambda I)^{-1} X_{\text{tr}}^\top \mathbf{y}_{\text{tr}}, \quad (2)$$

where $\mathbf{x}_b$ stacks log realized-volatility components at one-hour, four-hour and five-day scales, the current absolute return, intraday-bar dummies, and index dummies. The corrector is a low-rank

gated linear map on $\mathbf{x} = [\mathbf{x}_b; \text{nll}; \text{ent}; \mathbf{h}] \in \mathbb{R}^{556}$:

$$\Delta\hat{y} = \mathbf{w}_o^\top (\sigma(W_g \mathbf{x}) \odot W_r \mathbf{x}) + b, \quad \hat{y}^{M2} = \hat{y}^0 + \Delta\hat{y}, \quad (3)$$

with rank 32 (about 35K parameters), trained on baseline residuals with Adam and validation early stopping. Coefficients are frozen at the training boundary; an annual walk-forward refit changes out-of-sample R^2 by at most ± 0.005 over three years.

3.3 Evaluation metrics

Beyond incremental R^2 , we track four statistics. The *daily rank IC* of the incremental signal is the within-day Spearman correlation between the corrector’s increment and the baseline residual,

$$\text{IC}_d = \rho^S(\Delta\hat{y}_t, y_t - \hat{y}_t^0)_{t \in d}, \quad \text{IR} = \frac{\overline{\text{IC}}}{\text{sd}(\text{IC})}, \quad t = \text{IR} \sqrt{n_d}. \quad (4)$$

Forecast losses use QLIKE on variance forecasts with a train-sample multiplicative bias correction (Patton, 2011),

$$\text{QLIKE}(F_t, RV_t) = \frac{RV_t}{F_t} - \log \frac{RV_t}{F_t} - 1, \quad F_t = (c e^{\hat{y}_t})^2, \quad c = \exp(\overline{y - \hat{y}})_{\text{tr}}. \quad (5)$$

Loss comparisons aggregate to daily means and use the Diebold and Mariano (1995) statistic with a Newey–West variance (lag 5). Because the baseline is nested in the corrected model, we also report the Clark and West (2007) adjusted differential

$$f_t = e_{0,t}^2 - e_{1,t}^2 + (\hat{y}_t^0 - \hat{y}_t^{M2})^2, \quad (6)$$

whose daily mean is tested in the same way. Model comparisons across the full specification set use the model confidence set of Hansen et al. (2011) on daily QLIKE losses with a stationary block bootstrap.

4 Baseline Forecasting Results

Table 1 and Figure 1 contain the headline result. The scalar interface reproduces the historical null (+0.003–0.005); the full hidden state through the gated corrector delivers $\Delta R^2 = +0.065/+0.042/+0.096$ on validation/test/holdout, with a 100-seed holdout mean of $+0.095 \pm 0.004$ (minimum seed +0.084, none negative). LGBM applied to the same features confirms the attribution: without the hidden state it recovers only +0.013–0.019; with it, +0.035–0.079. A tree stacked after the gated corrector

Table 1: One-hour-ahead volatility: incremental R^2 over the strong baseline. The dependent variable is log mean absolute 5-minute return over the next hour. The baseline is a ridge regression on multi-scale realized volatility, the current absolute return, intraday-bar dummies, and index dummies, with coefficients frozen at the training boundary. The holdout (2025-07-18 to 2026-08-14) is pre-registered and opened once. The daily IC is the within-day Spearman rank correlation between the incremental prediction and the baseline residual; the daily IR is its mean over its standard deviation (a signal-strength ratio independent of sample length), and t multiplies the IR by $\sqrt{n_d}$.

Model	Validation	Test	Holdout	Holdout daily IC (IR; t)
Scalars (NLL + entropy)	+0.003	+0.005	—	—
Ridge + hidden state	+0.054	+0.034	+0.084	—
LGBM residual (no hidden)	+0.018	+0.013	+0.019	+0.098 (0.37; 5.9)
LGBM residual (+ hidden)	+0.055	+0.035	+0.079	+0.292 (1.04; 16.9)
GatedLinear (+ hidden)	+0.065	+0.042	+0.096	+0.377 (1.41; 22.9)

adds at most 0.002: with a strong baseline, the classical residual stage of [Dewage et al. \(2026\)](#) has nothing left to learn.

Statistical validation. QLIKE improves by 20.8%, 16.6%, and 20.5% across the three out-of-sample segments (Diebold–Mariano t statistics of 6.0, 5.7, and 7.6 on daily losses with Newey–West lag 5); Clark–West statistics for the nested comparison reach $t = 16.9$ on the holdout. A model confidence set over daily QLIKE losses (stationary bootstrap) eliminates every specification without hidden states at $p \leq 0.021$ in both frozen segments. Shuffling hidden states across days within the same intraday bar collapses the increment to +0.005–0.009—the scalar level. Regressing the incremental prediction on all baseline variables plus the scalars yields out-of-sample $R^2 \approx 0$: the signal is essentially one hundred percent novel. Finally, the gain is largest exactly when the baseline struggles: the daily rank correlation between the corrector’s mean-squared-error gain and the baseline’s mean-squared error is 0.55–0.64 in every segment, and the learned gate opens monotonically in the realized-volatility quintile. Table 2 collects these statistics; Figure A1 shows the full 100-seed distribution, Figure A2 the daily IC through time, and Figures A3–A4 the two mechanism facts: gains concentrate on days where the baseline errs most, and the gate opens with realized volatility.

5 The Nature of the Extracted Information

Direction: nothing. Every configuration—scalars, ridge, gated corrector, and a 512-dimensional ridge on returns—yields $R^2 \leq 0.003$ for one-hour returns. Window-local normalization strips level information; what remains is shape, and shape surprise is a second-moment quantity. This doubles as a leakage control, since any look-ahead would inflate the directional target as well.

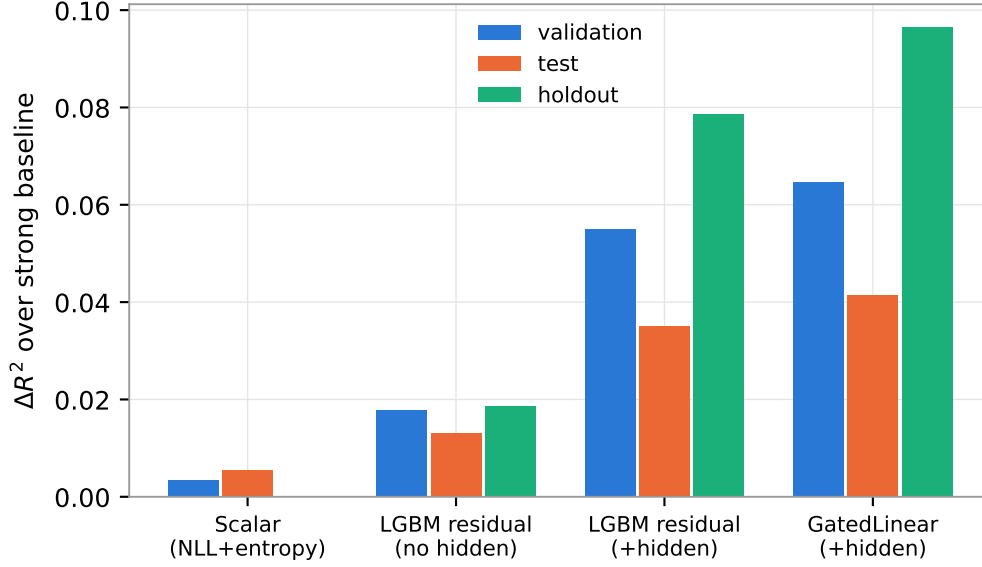


Figure 1: The extraction interface is the bottleneck. Incremental R^2 over the strong baseline for four extraction interfaces applied to the same frozen model, by out-of-sample segment. The scalar interface sits in the historical “dead zone”; the full hidden state is an order of magnitude larger, and the gated corrector dominates tree-based extraction.

Horizon: an inverted-U peaking at one hour. Sweeping the forecast horizon from one bar to the next day traces a sharply humped profile of the hidden-state increment (Figure 3): +0.013 at five minutes, +0.047 at fifteen, +0.066 at thirty, +0.084 at one hour, +0.075 at two hours, and +0.009 for next-day realized variance over a Log-HAR baseline, where the gated corrector in fact *overfits* at the daily sample size (holdout -0.046). The representation therefore carries information at a characteristic timescale of roughly one hour: it is neither tick-level noise nor an overnight-persistent state variable. This single curve organizes the application results in Section 7—it predicts, correctly, both where the forecast can be monetized and where it cannot.

The forecast subsumes the option market’s own volatility expectation. A sharper benchmark than any statistical model is what the derivatives market itself expects. We extract at-the-money straddle-implied volatility for CSI 1000 (MO) index options at five-minute frequency from the full order book over 133 trading days, all of which fall inside the pre-registered holdout window, so the corrector’s coefficients have never seen them. Implied volatility alone explains $R^2 = 0.19$ of intra-day realized volatility and adds only +0.012 over our baseline; adding the hidden-state correction lifts R^2 from 0.505 to 0.654. The bidirectional test is decisive: conditioning on implied volatility, the hidden-state increment retains a daily rank IC of +0.405 (daily IR 1.20, positive on 87% of days), whereas conditioning on the hidden state, implied volatility’s own increment turns slightly negative (-0.086). Consistent with the variance risk premium literature (Carr and Wu, 2009), implied volatility exceeds subsequent realized volatility by 0.43 log units in our sample. Two caveats

Table 2: Statistical validation of the forecast gain. QLIKE is computed on variance forecasts with a train-sample multiplicative bias correction; DM statistics use daily mean loss differentials with Newey–West lag 5; CW is the [Clark and West \(2007\)](#) statistic for nested models. The model confidence set uses daily QLIKE losses and a stationary block bootstrap.

	Validation	Test	Holdout
QLIKE improvement	−20.8%	−16.6%	− 20.5%
Diebold–Mariano t	5.96	5.67	7.55
Clark–West t	13.40	12.07	16.93
MCS survivors ($\alpha = 0.10$)	—	{M2, M3}	{B3, M2, M3}
Placebo (shuffled hidden) ΔR^2	+0.009	+0.005	+0.008
100-seed ΔR^2 (mean $\pm$ sd)	+0.065 $\pm$.002	+0.045 $\pm$.003	+0.095 $\pm$.004

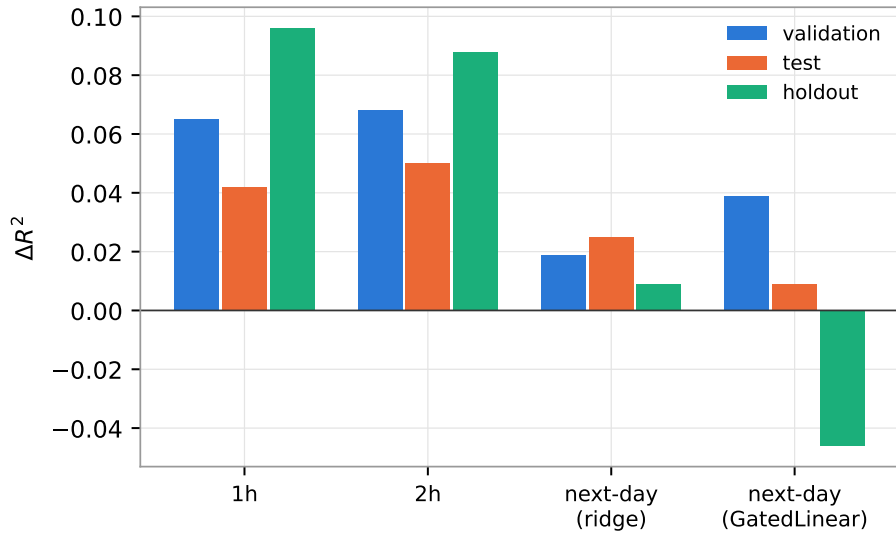


Figure 2: Term structure of the hidden-state increment. The information horizon is strictly intraday. The daily-frequency gated corrector illustrates validation-selected overfitting that only the pre-registered holdout exposes.

bound the claim: listed options price a monthly horizon while we forecast one hour, and no hourly volatility instrument exists in this market—the informational advantage is real but currently lacks a carrier.

Tails: symmetric only. The 95th percentile of future absolute returns inherits the full gain (+0.033–0.070); signed value-at-risk pinball losses improve by only ± 2 –6% with mixed signs, and crash-probability AUC moves by ± 0.01 .

Subsumption of cross-sectional breadth. Against a panel of four cross-sectional signals built from all 500 index constituents (breadth entropy, breadth NLL, close-location and upper-wick breadth), the hidden-state increment is undiminished (+0.053–0.058), while the breadth increment conditional on the hidden state is +0.001: one index series passed through a frozen TSFM contains

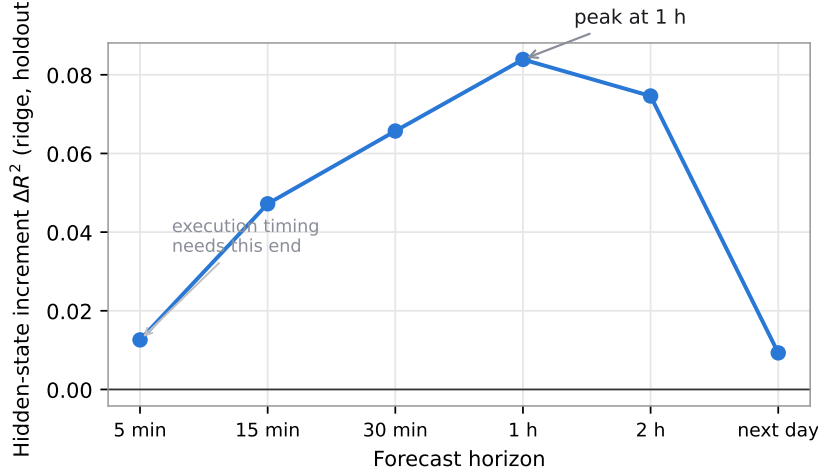


Figure 3: The hidden-state increment has a characteristic timescale. Incremental R^2 from adding the frozen model’s hidden state to the baseline, by forecast horizon, on the pre-registered holdout (ridge specification, so all horizons are compared under an identical estimator). The profile peaks at one hour. Execution scheduling requires accuracy at the left end, where the increment is smallest—which is why it fails (Section 7).

the volatility-relevant content of the 500-stock cross-section.

6 Is Pretraining Necessary?

6.1 Scratch networks and data efficiency

A scratch Transformer (dimension 64, two layers) trained end-to-end on the identical normalized window—without the model-derived scalars—matches the frozen pipeline on validation and holdout at full sample size (+0.065/+0.035/+0.096), trailing only on the test segment; a scratch GRU and a PCA-512 control fall short everywhere. The decisive comparison is data efficiency (Figure 4): restricted to the most recent 50% of training days (about 4.5 years), the frozen representation retains +0.074 on the holdout (77% of its full-sample gain) versus +0.033 (34%) for the scratch model; at 15% the ordering is the same; at 5% both fail, with the high-dimensional corrector failing worse. The premium of the pretrained representation is concentrated precisely in the data regime practitioners occupy.

6.2 Representation scale, model class, and extraction views

Kronos-base (four times the parameters) adds only +0.002–0.008 by ridge; the small model saturates this task. A univariate Chronos-T5-small encoder on close prices alone yields a real but three-times-smaller increment (+0.056 on holdout by ridge): the phenomenon is generic to TSFMs, while

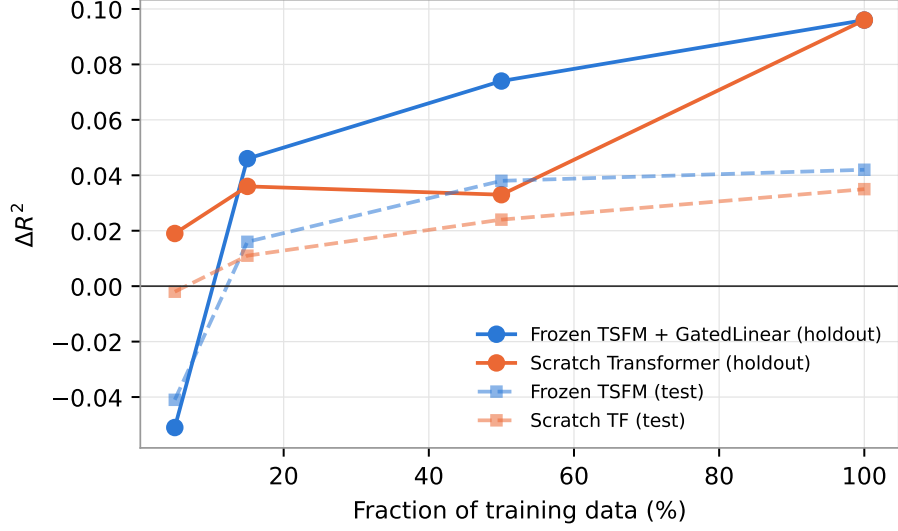


Figure 4: Data efficiency: the economic case for pretraining. Incremental R^2 as a function of the fraction of training data (most recent days retained; both arms share the correspondingly re-fit baseline). The frozen representation reaches most of its full-sample gain with half the data; the scratch Transformer requires all of it.

multivariate OHLCV input and K-line-domain pretraining explain Kronos’s advantage. Multi-layer and multi-position extraction (four concatenated views, 2,048 dimensions) markedly improves the *linear* interface (test +0.034 $\rightarrow$ +0.051) but does not surpass the production single-view gated corrector: the nonlinear corrector already performs the cross-position aggregation internally.

6.3 Transfer

Within the broad-cap family, a corrector trained on CSI 300 and CSI 1000 only transfers to CSI 500 at full strength (+0.055/+0.063/+0.102). Zero-shot transfer to style indices (SSE 50, ChiNext, STAR 50) retains only +0.001–0.012, but re-fitting the 35K-parameter corrector locally on the frozen representation recovers +0.023–0.062: the representation is universal; the corrector mapping is family-specific. Figure 5 summarizes both experiments.

7 Economic Application: A Futures Overlay under Realistic Costs

We map forecasts to positions by volatility targeting with a trend gate and a no-trade band,

$$w_t = 1 + \left[\text{clip}(\sigma^*/\hat{\sigma}_t, 1, 3) - 1 \right] \cdot \mathbf{1}\{\text{mom}_{k,t-1} > 0\}, \quad \sigma^* = \text{median}_{\text{tr}}(\hat{\sigma}), \quad (7)$$

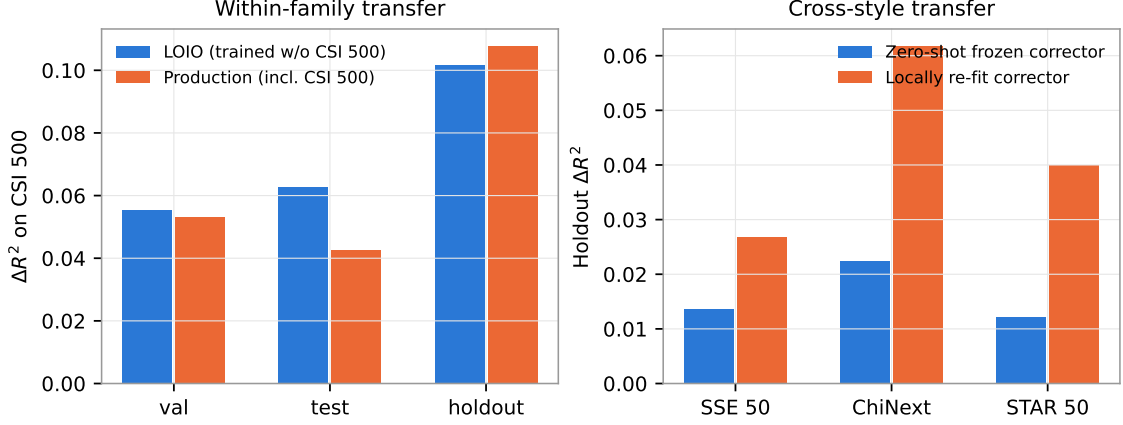


Figure 5: Transfer of the corrector. Left: leave-one-index-out training within the broad-cap family transfers to CSI 500 at full strength. Right: zero-shot transfer of the frozen corrector to style indices retains only a fraction of the gain (holdout shown), while re-fitting the 35K-parameter corrector locally on the frozen representation restores most of it.

executing a change only when $|w_t - w^{\text{cur}}| \geq \delta = 0.3$; all hyperparameters are chosen on train-plus-validation and frozen. Daily strategy returns follow close-to-close accounting with within-contract futures returns r_t and the asymmetric Chinese fee schedule,

$$R_d = \sum_{t \in d} w_{t-1} r_t - c^+ (\Delta w_t)_+ - c^- (\Delta w_t)_- - c_t^{\text{roll}}, \quad c^+ = 0.55 \text{ bp}, \quad c^- = 3.77 \text{ bp}, \quad (8)$$

where intraday close-outs cost roughly seven times more than opens and day-first reductions and rolls are charged at the overnight rate. The no-trade band cuts daily turnover from 1.4 to 0.4 and is what keeps the overlay alive under the close-out fee.

Replacing the baseline forecast with the hidden-state forecast in an otherwise identical overlay is worth +450 to +971 basis points per year on IC and +464 to +690 on IM (six of six index-segment cells positive; best cell $t = 2.3$; Figure 7); the break-even symmetric cost is approximately 3 basis points (Figure 6). Table 3 summarizes the overlay economics. Against passively holding the (discounted) futures, the gated overlay is positive in all four frozen cells (+162–606 basis points) but with $t < 1.5$, and stationary-bootstrap 95% intervals for both comparisons include zero. We report this gap deliberately: forecast improvements that are overwhelming by Clark–West standards can still be marginal at the strategy level once carry, overnight gaps, and costs enter. The deployment we endorse is an overlay for accounts that already hold a *net long* index-futures position: applying the same rule to a short hedging leg reverses the sign of the gain (the increment comes from leveraging up in low-volatility states, which coincide with positive drift), so the overlay is not a general position-scaling device. We also note that the scaling exponent is a regime bet: $1/\sigma^2$ scaling (as in [Moreira and Muir, 2017](#)) outperforms in the calm holdout year and underperforms sharply in the test year,

Table 3: Futures overlay economics. The left block compares volatility-targeted overlays driven by the hidden-state forecast (V2) versus the baseline forecast (V1) in identical construction (symmetric one-basis-point cost); annualized return differences with Newey–West t statistics in parentheses. The right block compares the frozen gated overlay (trend gate, no-trade band, realistic asymmetric costs) with passively holding the dominant contract. All configuration choices use train-plus-validation data only.

Product	Segment	Forecast upgrade (V2–V1)		Overlay vs. buy-and-hold	
		bp/year	(t)	Sharpe	BH Sharpe
IC	Validation	+450	(1.39)	—	—
IC	Test	+63	(0.17)	0.94	0.88
IC	Holdout	+971	(2.28)	1.42	1.36
IM	Validation	+690	(1.81)	—	—
IM	Test	+483	(1.23)	1.13	1.15
IM	Holdout	+464	(1.00)	1.29	1.17

whereas the forecast upgrade improves the $1/\sigma$ overlay in all four cells.

7.1 The position map, not the forecast, is the binding constraint

Comparing our overlay against a textbook rule driven by one-day trailing realized volatility exposes where the forecast advantage is lost. Under the production rule with realistic costs the two are statistically indistinguishable, despite a holdout forecasting gap of $R^2 = 0.493$ versus 0.634. Two mechanisms attenuate the signal. First, *truncation*: the map $w = \text{clip}(\sigma^*/\hat{\sigma}, 1, 3)$ pins 79% of observations at unit exposure, so most forecast variation never reaches the portfolio. Second, *turnover*: a more responsive forecast trades more, and intraday close-outs are penalized sevenfold.

Replacing the truncating map with a smooth one confirms the diagnosis. With the mapping family selected on train-plus-validation and frozen thereafter, truncating maps deliver negative increments over the textbook rule in all four frozen product-segment cells (−276 to −689 bp per year), while exponentially smoothed and soft-power maps deliver positive increments in all of them (+7 to +976 bp). Smoothing preserves the ordering of the forecast while controlling turnover; truncation discards the ordering outright.

This is precisely the structure implied by [Gârleanu and Pedersen \(2013\)](#), whose framework solves for the position path when returns are predictable and trading is costly. Its solution takes two forms: with proportional costs the optimum is a no-trade band—exactly the δ rule we arrived at empirically—while with quadratic costs it is partial adjustment toward an aim portfolio, $w_t = (1 - a)w_{t-1} + a \text{aim}_t$, which is exponential smoothing. We estimate all inputs from the data rather than tuning them: the aim portfolio’s mean-reversion rate ($\phi = 0.0216$ per bar, a half-life of 32 bars ≈ 2.7 hours, itself consistent with the one-hour peak of Figure 3), the per-bar return variance, and a quadratic cost coefficient matched to the fee schedule at the typical trade size. The resulting

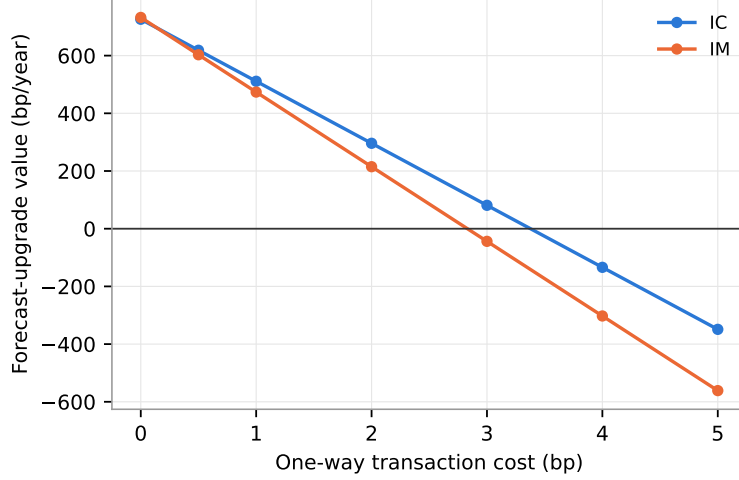


Figure 6: Cost sensitivity of the forecast upgrade. Annualized value of replacing the baseline forecast with the hidden-state forecast in an identical overlay, as a function of the symmetric one-way transaction cost (test and holdout combined). The break-even cost is approximately three basis points for both contracts.

trading rate,

$$a^* = \frac{\sqrt{(G + \lambda\rho)^2 + 4G\lambda} - (G + \lambda\rho)}{2\lambda} = 0.302, \quad G \equiv \gamma\Sigma, \quad (9)$$

is obtained with *zero* tuning and delivers holdout Sharpe ratios of 2.03 (IC) and 1.59 (IM), matching the grid-searched smoothing constant (2.02 and 1.68) while consuming no out-of-sample degrees of freedom. The implied shrinkage of the aim portfolio, $G/(G + a^*\lambda\phi) = 0.95$, is mild, as it should be for a signal whose half-life exceeds the rebalancing interval by two orders of magnitude. That the empirically superior map is smoothing rather than banding suggests effective costs here are dominated by impact (quadratic) rather than fees (proportional).

7.2 Where the forecast cannot be monetized

The horizon curve also predicts a failure. Execution scheduling—choosing which bars to trade a fixed quantity in, which incurs no incremental turnover—requires accuracy at the one-bar horizon, precisely where the hidden-state increment is smallest (+0.013). Under a causal participation schedule, the baseline forecast reduces volume-weighted realized volatility by 3.85% relative to a uniform schedule and the textbook rule by 1.04%, but the hidden-state forecast by only 2.97%: adding a horizon-mismatched signal makes execution worse, not better. Cost-aware end-to-end training fails in both variants we tried—restricting the loss to the interior of the position map collapses forecast accuracy, and optimizing net profit directly halves holdout Sharpe—because return realizations are too noisy to supervise a 35K-parameter map. At this signal-to-noise ratio, forecasting accurately and then using the forecast rationally dominates learning the decision end to end.

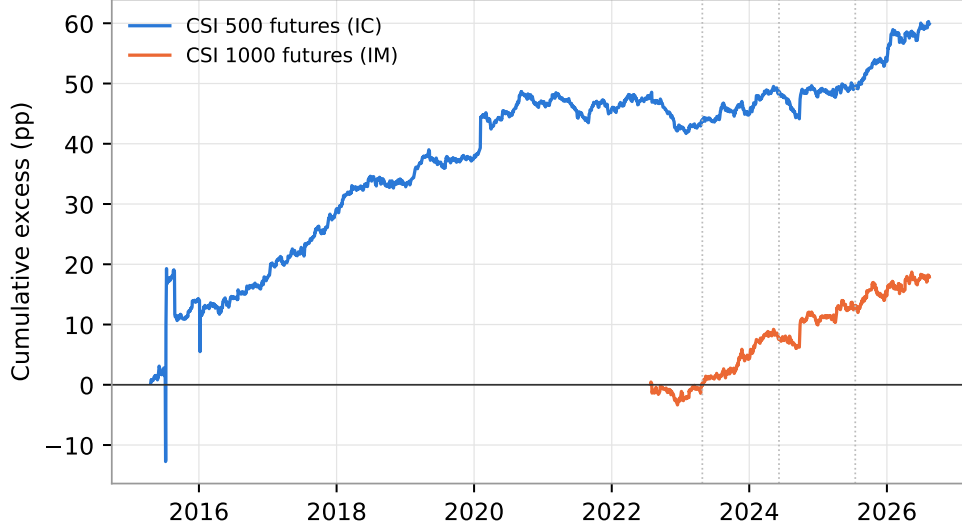


Figure 7: Cumulative value of the forecast upgrade on index futures. Cumulative daily return difference between volatility-targeted overlays driven by the hidden-state forecast versus the baseline forecast (identical construction, one-basis-point cost), with within-contract returns, overnight gaps, and roll accounting. Dotted lines mark the validation/test/holdout boundaries.

8 Robustness

Our conclusions survive the following checks. (i) One hundred re-trainings from independent seeds: holdout $\Delta R^2 = +0.095 \pm 0.004$ with no negative seed. (ii) A placebo that shuffles hidden states across days within the same intraday bar collapses the gain to the scalar level. (iii) Annual walk-forward refits versus frozen coefficients differ by at most ± 0.005 over three years. (iv) All sixteen cells of the overlay hyperparameter grid are positive in both frozen segments, and the selected cell lies *below* the grid median on the holdout, so reported overlay numbers are conservative. (v) Every selection uses train-plus-validation data only. (vi) A complete no-look-ahead audit (twelve design checks and six empirical checks) accompanies the code, including the one-day lag of the dominant-contract map and the within-contract construction of futures returns.

Limitations. The forecast advantage does not translate one-for-one into the strategy application. Under the production rule with realistic costs, a textbook overlay driven by a one-day trailing realized volatility attains Sharpe ratios of 0.92/1.63 (IC) and 1.13/1.32 (IM) on test/holdout versus 0.94/1.41 and 1.16/1.26 for the hidden-state forecast—a tie, despite a holdout forecasting gap of $R^2 = 0.493$ versus 0.634. Two mechanisms explain the attenuation: the position map is truncated at unit exposure for 79% of observations, discarding most forecast variation, and the smoother trailing estimator turns over far less under the punitive close-out fee. Our economic claims are therefore stated strictly as V2 versus V1 (same forecast family), and we do not claim superiority over textbook volatility targeting. Fine-tuning (e.g., LoRA) is untested for lack of GPU access; results concern

one market and a family of related indices with commercial minute data; the impact-cost assumption (0.32 basis points) has not been validated against order-book data; and the corrector requires local re-fitting outside the broad-cap index family.

9 Conclusion

The unit of evaluation for foundation models in finance should be the representation, not the model's scalar summaries. A frozen TSFM's hidden state carries intraday second-moment information that scalar interfaces discard, that a 500-stock cross-section does not add to, and that a scratch network can only match with years of additional data. The practical recipe is small: one forward pass per decision point, a 35K-parameter gated corrector, and honest accounting on the way to the futures market.

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Appendix: Additional Figures

This appendix collects supporting robustness and mechanism figures: the 100-seed distribution (Figure A1), the daily IC through time (Figure A2), and the two mechanism facts (Figures A3 and A4).

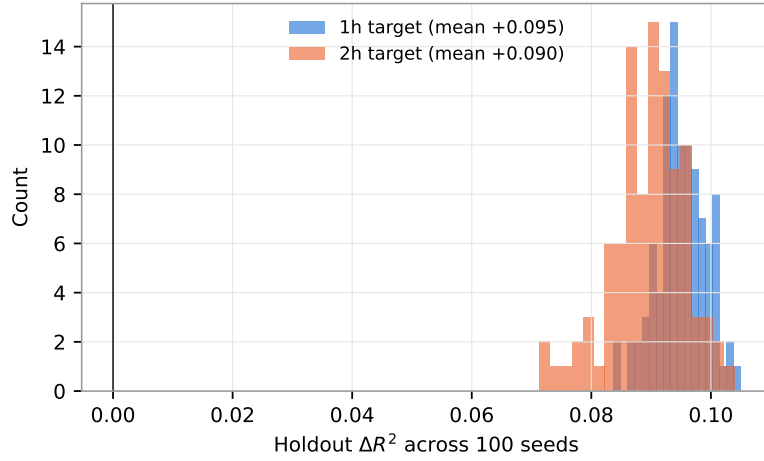


Figure A1: Seed robustness. Distribution of the holdout ΔR^2 across 100 independent re-trainings of the gated corrector for the one-hour and two-hour targets. No seed is negative for either target.

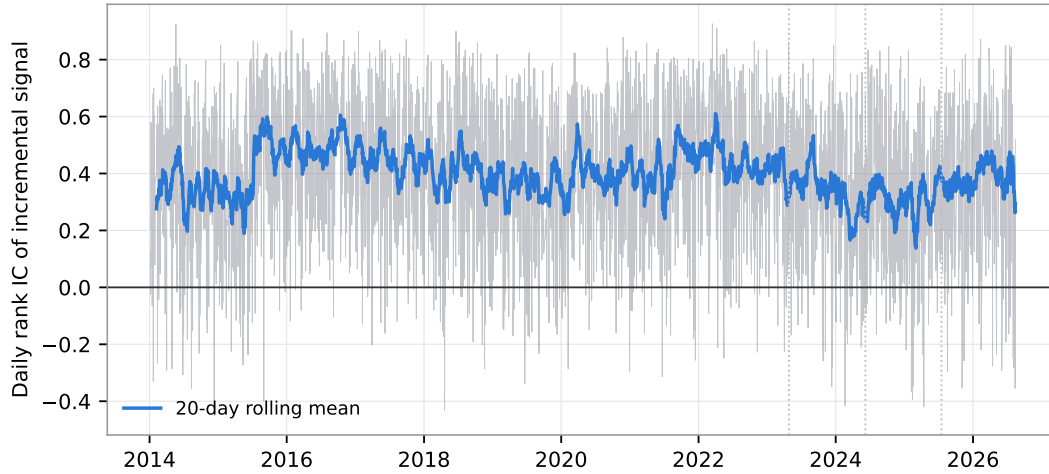


Figure A2: Daily rank IC of the incremental signal through time. Thin line: daily values; thick line: 20-day rolling mean. Dotted verticals mark the validation/test/holdout boundaries. The signal does not decay out of sample: among the three out-of-sample segments, the mean daily IC is highest in the pre-registered holdout year (0.38 versus 0.34 and 0.31).

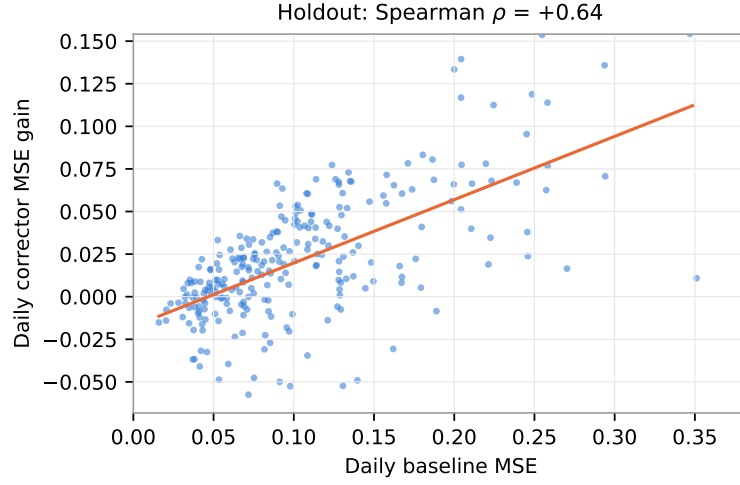


Figure A3: Gains concentrate where the baseline fails. Each dot is one holdout day: daily baseline mean-squared error against the corrector’s daily MSE gain, with an OLS fit. The Spearman correlation is $+0.64$ on the holdout ($+0.57$ and $+0.55$ on validation and test).

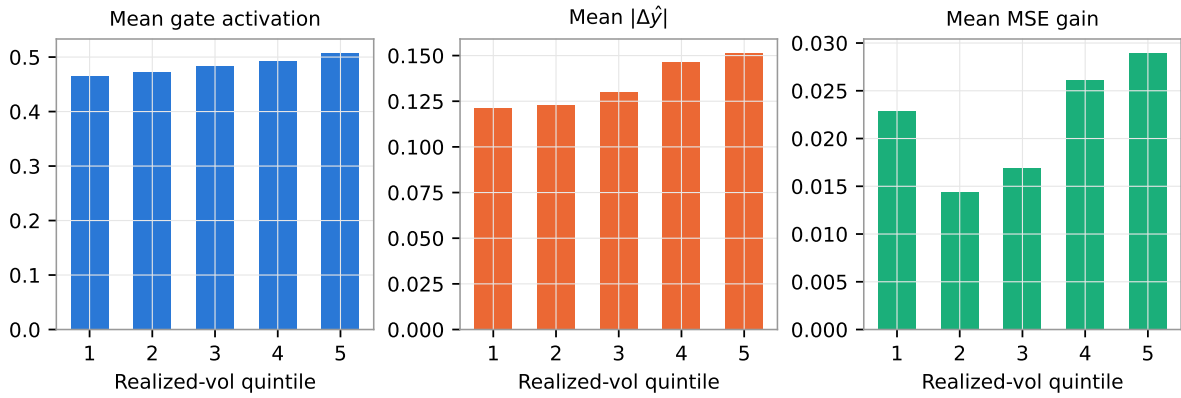


Figure A4: The learned gate opens with volatility. Holdout means of the gate activation $\sigma(W_g \mathbf{x})$, the absolute correction $|\Delta\hat{y}|$, and the realized MSE gain, by train-sample realized-volatility quintile.